DAY 04 ASSIGNMENT

UNDERSTANDING TCP/IP & OSI MODEL

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Assignment Code: DAY04-TCPIP-OSI
Topic: TCP/IP and OSI Model

1) OSI/TCP-IP THEORY

1) What is a network protocol?

A network protocol is a set of rules that enable computers and other devices to communicate with each other. An example of some common network protocols would be TCP, IP, HTTP, HTTPS and DNS.

2) What is the OSI Model?

The Open Systems Interconnection model is what is known as the reference model. It explains how data communicates between devices on a network and contains 7 layers. Each layer has a specific purpose.

3) What are the 7 OSI layers in order?

1. Application Layer (Layer 7)
2. Presentation Layer (Layer 6)
3. Session Layer (Layer 5)
4. Transport Layer (Layer 4)
5. Network Layer (Layer 3)
6. Data Link Layer (Layer 2)
7. Physical Layer (Layer 1)

4) What does the Application Layer do?

This is the top layer of the OSI Model. The Application Layer provides network services for applications. Users can access services such as websites, email and file transfer. Protocols at this layer include HTTP, HTTPS, FTP, SMTP and DNS.

5) What does the Transport Layer do?

This layer handles communication between applications on different devices. The data is split into segments and port numbers are used to identify applications and services. TCP provides reliable communication and UDP provides fast communication.

6) What does the Network Layer do?

This layer handles the logical addressing of devices and the routing of data between different networks. IP Addresses are used to identify the source and destination devices. Routers can use this information to help route packets to their destination. IP is the key protocol used at this layer.

7) What does the Data Link Layer do?

This layer is responsible for transferring data between devices on the local network. Data is organised into frames and MAC Addresses are used to identify devices on the local network. Ethernet and Wi-Fi operate at this layer.

8) What does the Physical Layer do?

This is the bottom layer of the OSI model. It handles the transmission of unprocessed bits of data across a communication medium. Cables, connectors and signals (electrical, light or wireless) are all part of this layer.

9) What is the TCP/IP Model?

The Transmission Control Protocol/Internet Protocol Model is used to explain how devices communicate over networks. This is often made up of 4 layers. These are the Application, Transport, Internet and Network Access. Each layer has a different purpose to help data travel from one device to another.

10) What is the difference between the OSI and TCP/IP Models?

OSI Model vs TCP/IP Model

OSI Model has 7 layers whereas the commonly used TCP/IP Model has 4 layers.

The OSI Model is a reference model however the TCP/IP Model is based around the protocols used.

The Application, Presentation and Session Layers are separate in OSI Model whereas these are merged together in the TCP/IP Model under the Application Layer.

11) What is Encapsulation?

Encapsulation is when headers are added to data as it goes through the layers of the network. Each layer will add the required data to help it communicate. The Transport Layer will add TCP or UDP headers and the Network Layer will add IP headers. This data is now ready to be sent.

12) What is Decapsulation?

Decapsulation is the opposite of encapsulation. Each layer of the network will remove and process the relevant headers. The data will then be passed to the receiving application.

13) Which layer uses IP Addresses?

IP Addresses are used at Layer 3 (Network Layer) of the OSI Model. This helps identify the source and destination of data packets. Routers can use this information to help route data between networks.

14) Which layer uses MAC Addresses?

MAC Addresses are used at Layer 2 (Data Link Layer). These help identify interfaces on a network.

15) Which layer uses TCP & UDP?

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are used at Layer 4 (Transport Layer). TCP offers reliable communication whereas UDP offers unreliable communication.

16) Which layer contains HTTP & HTTPS?

HTTP and HTTPS operate at Layer 7 (Application Layer) of the OSI Model. These are used for communication between browsers and servers. HTTP is used to send web content whereas HTTPS is used to securely communicate using encryption via TLS.

17) Why should Cybersecurity Professionals know about network layers?

As a Cybersecurity Professional, it is important to know about network layers. They provide an understanding of how data travels between devices and networks. Each layer has a different purpose and there could be issues with security. Security Analysts will understand how to identify suspicious traffic, detect attacks on networks, investigate unwanted

connections and understand how firewalls and security tools operate. They will also understand how to analyse packets on networks and where any issues with security may occur.

18) How does data travel across networks from the browser to the server?

When a user types in the address of a website, the browser will need to find the IP address of this website. If it cannot be found from the cache, the browser will use the DNS to translate the domain name into an IP address.

Once the browser has the IP address, it will connect to the web server. A 3 way handshake will happen using TCP (as this is a typical connection using HTTPS) and then a TLS handshake.

The browser will send an HTTP request to the server. Data is then sent through the network layers. The Transport Layer will split the data up into segments. The Network Layer will add IP addressing information. The Data Link Layer will create frames with MAC addresses.

The data will be sent through the network using signals at the Physical Layer. Routers will route the packets to the destination using the IP address. At each network link, data will be prepared for that link.

Once the data has reached the server, Decapsulation will occur. At each layer, headers will be processed and removed. The data will then be sent to the application on the server. The server will process the request and respond back to the browser.

2) PRACTICAL

19) Resolved IP Address for Google.com using nslookup.

I used the nslookup command on the Google.com domain in the Windows Command Prompt.

Resolved IP Address: 142.251.221.206

20) Number of hops to Google.com using traceroute.

I used the tracert command on the Google.com domain to identify the number of hops.

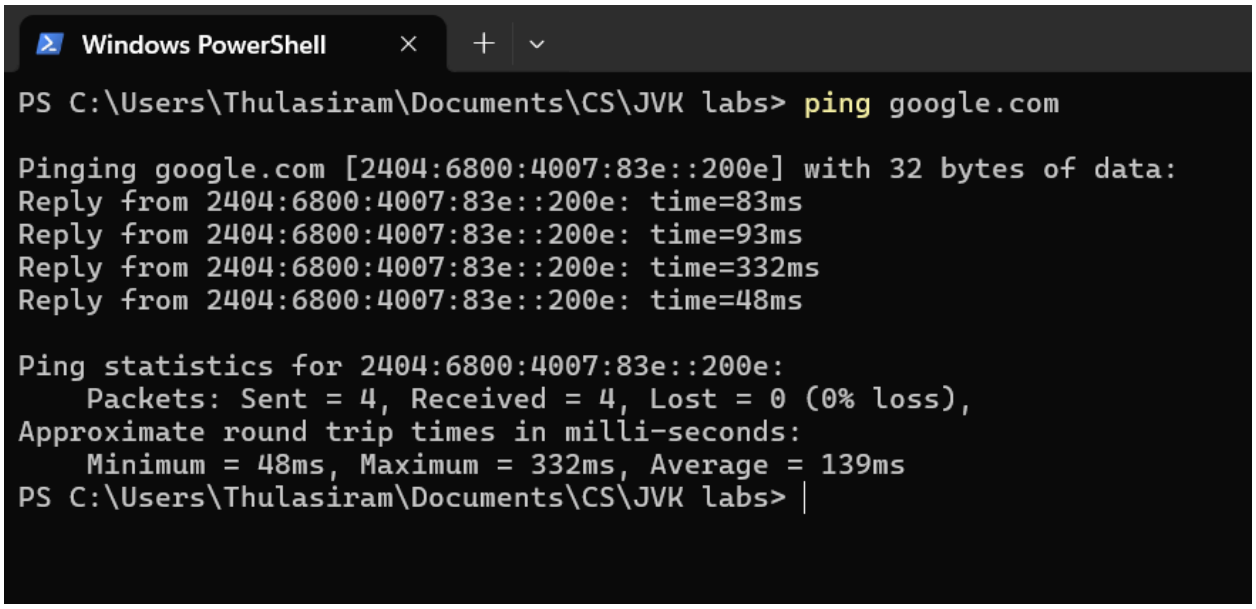
Number of Hops: 12

[Note: The number of hops may vary depending on the network. If the destination was not reached, I will note this and state which hops responded.]

3) SCREENSHOTS

Screenshot 1: Ping Google.com

This is the output from the ping command run in the Command Prompt on Google.com. It shows the resolved IP address, responses from Google and an overview of the packets sent.



Screenshot 2: Tracert Google.com

This is the output from the tracert command run on Google.com. It shows the hops across the network.

```
PS C:\Users\Thulasiram> tracert -4 google.com

Tracing route to google.com [142.251.221.206]
over a maximum of 30 hops:

  1     3 ms     6 ms     3 ms  10.47.230.60
  2    25 ms     6 ms     4 ms  192.0.0.1
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12    85 ms    78 ms    72 ms  pnmaaa-ba-in-f14.1e100.net [142.251.221.206]

Trace complete.
```

Screenshot 3: Nslookup Google.com

```
Windows PowerShell
PS C:\Users\Thulasiram> nslookup google.com
Server:  UnKnown
Address:  10.47.230.60

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:831::200e
          142.251.221.206

PS C:\Users\Thulasiram>
```

This is the output from the nslookup command run on Google.com. It shows the information provided by the DNS Server and the IP address(es).

CONCLUSION

By completing this assignment I have learnt about network protocols, the OSI Model and TCP/IP Model. I now understand how the 7 layers of the OSI Model operate and what their purpose is. I have also learnt about encapsulation, decapsulation, IP Addresses, MAC Addresses, TCP, UDP, HTTP and HTTPS. I have learnt how to use the ping, tracert and nslookup

commands to help me identify network connectivity, network routes and domain names to IP Addresses. This knowledge is useful as it can help me to understand network communication and how Cybersecurity Professionals analyse networks.